

0902.0431: table evidence

Tables

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_TAB_001	17	—	$\begin{tabular}{t l l l l l l l } \hline & \text{\(S_1\)} \\ & \text{\(S_2\)} & \text{\(S_3\)} & \text{\(S_4\)} & \text{\(S_5\)} & \text{\(S_6\)} \\ \hline H_1 & \text{\(S_2\)} & \text{\(-S_1\)} & \text{\(-S_4\)} & \text{\(S_3\)} & \text{\(0\)} & \text{\(0\)} \\ H_2 & \text{\(0\)} & \text{\(0\)} & \text{\(S_4\)} & \text{\(-S_3\)} & \text{\(-S_6\)} & \text{\(S_5\)} \\ L_{12} & \text{\(-S_3\)} & \text{\(-S_4\)} & \text{\(S_1\)} & \text{\(S_2\)} & \text{\(0\)} & \text{\(0\)} \\ L_{21} & \text{\(S_4\)} & \text{\(-S_3\)} & \text{\(S_2\)} & \text{\(-S_1\)} & \text{\(0\)} & \text{\(0\)} \\ L_{13} & \text{\(-S_5\)} & \text{\(-S_6\)} & \text{\(0\)} & \text{\(0\)} & \text{\(S_1\)} & \text{\(S_2\)} \\ L_{31} & \text{\(S_6\)} & \text{\(-S_5\)} & \text{\(0\)} & \text{\(0\)} & \text{\(S_2\)} & \text{\(-S_1\)} \\ L_{23} & \text{\(0\)} & \text{\(0\)} & \text{\(-S_5\)} & \text{\(-S_6\)} & \text{\(S_3\)} & \text{\(S_4\)} \\ L_{32} & \text{\(0\)} & \text{\(0\)} & \text{\(S_6\)} & \text{\(-S_5\)} & \text{\(S_4\)} & \text{\(-S_3\)} \\ \hline \end{tabular}$	$\begin{array}{c cccccc} & S_1 & S_2 & S_3 & S_4 & S_5 & S_6 \\ \hline H_1 & S_2 & -S_1 & -S_4 & S_3 & 0 & 0 \\ H_2 & 0 & 0 & S_4 & -S_3 & -S_6 & S_5 \\ L_{12} & -S_3 & -S_4 & S_1 & S_2 & 0 & 0 \\ L_{21} & S_4 & -S_3 & S_2 & -S_1 & 0 & 0 \\ L_{13} & -S_5 & -S_6 & 0 & 0 & S_1 & S_2 \\ L_{31} & S_6 & -S_5 & 0 & 0 & S_2 & -S_1 \\ L_{23} & 0 & 0 & -S_5 & -S_6 & S_3 & S_4 \\ L_{32} & 0 & 0 & S_6 & -S_5 & S_4 & -S_3 \end{array}$	$\begin{array}{c cccccc} & S_1 & S_2 & S_3 & S_4 & S_5 & S_6 \\ \hline H_1 & S_2 & -S_1 & -S_4 & S_3 & 0 & 0 \\ H_2 & 0 & 0 & S_4 & -S_3 & -S_6 & S_5 \\ L_{12} & -S_3 & -S_4 & S_1 & S_2 & 0 & 0 \\ L_{21} & S_4 & -S_3 & S_2 & -S_1 & 0 & 0 \\ L_{13} & -S_5 & -S_6 & 0 & 0 & S_1 & S_2 \\ L_{31} & S_6 & -S_5 & 0 & 0 & S_2 & -S_1 \\ L_{23} & 0 & 0 & -S_5 & -S_6 & S_3 & S_4 \\ L_{32} & 0 & 0 & S_6 & -S_5 & S_4 & -S_3 \end{array}$
0902.0431_TAB_002	174	—	$\begin{tabular}{t l l l l l l l l } \hline & \text{\(\mu_1 - \frac{1}{3} \nu - r = 0 \)} & \text{\(1 \)} & \text{\(1 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} \\ \hline \mu_1 - \frac{1}{3} \nu + r = 0 & \text{\(0 \)} & \text{\(1 \)} & \text{\(1 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} \\ -\mu_3 + \frac{1}{3} \nu + r = 0 & \text{\(1 \)} & \text{\(2 \)} & \text{\(1 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(1 \)} \\ \mu_1 - \frac{1}{3} \nu - r = 0 & \text{\(1 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} \\ \mu_2 - \frac{1}{3} \nu + r = 0 & \text{\(0 \)} & \text{\(0 \)} & \text{\(1 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} \\ -\mu_3 - \frac{1}{3} \nu - r = 0 & \text{\(1 \)} & \text{\(1 \)} & \text{\(1 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(0 \)} & \text{\(1 \)} \\ \hline \end{tabular}$	$\begin{array}{c cccccc} \mu_1 - \frac{1}{3} \nu + r = 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ \mu_2 - \frac{1}{3} \nu + r = 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ -\mu_3 + \frac{1}{3} \nu + r = 0 & 1 & 2 & 1 & 0 & 0 & 0 & 1 \\ \mu_1 - \frac{1}{3} \nu - r = 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ \mu_2 - \frac{1}{3} \nu + r = 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -\mu_3 - \frac{1}{3} \nu - r = 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}$	$\begin{array}{c cccccc} \mu_1 - \frac{1}{3} \nu + r = 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ \mu_2 - \frac{1}{3} \nu + r = 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ -\mu_3 + \frac{1}{3} \nu + r = 0 & 1 & 2 & 1 & 0 & 0 & 0 & 1 \\ \mu_1 - \frac{1}{3} \nu - r = 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ \mu_2 - \frac{1}{3} \nu + r = 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ -\mu_3 - \frac{1}{3} \nu - r = 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}$

[illegible]

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_TAB_004	—	—	\begin{tabular}{l} Ichiro Yokota \end{tabular}	Ichiro Yokota	—
0902.0431_TAB_005	—	—	\begin{tabular}{lr} 1.1. Cayley algebra $\text{\es {C}}$ \\ \quad \quad \quad & \text{\cdots \;}; 1$ \vspace{1mm}\\ 1.2. Compact exceptional Lie group G_2$ \quad & \text{\cdots \;}; 2$ \vspace{1mm}\\ 1.3. Outer automorphism of the Lie algebra \text{\es {d}}_4$ \quad & \text{\cdots \;}; 3$ \vspace{1mm}\\ 1.4. Lie algebra \text{\es {g}}_2 of G_2$ \quad & \text{\cdots \;}; 8$ \vspace{1mm}\\ 1.5. Lie subalgebra \text{\es {su}}(3) of \text{\es {g}}_2$ \quad & \text{\cdots \;}; 9$ \vspace{1mm}\\ 1.6. Simplicity of \text{\es {g}}_2^C$ \quad & \text{\cdots \;}; 11$ \vspace{1mm}\\ 1.7. Killing form of \text{\es {g}}_2^C$ \quad & \text{\cdots \;}; 13$ \vspace{1mm}\\ 1.8. Roots of \text{\es {g}}_2^C$ \quad & \text{\cdots \;}; 14$ \vspace{1mm}\\ 1.9. Automorphism w of order 3 and subgroup SU(3) of G_2$ & \quad \text{\cdots \;}; 16$ \vspace{1mm}\\ 1.10. Involution \gamma and subgroup (Sp(1) \times Sp(1))/\text{\es {Z}}_2 of G_2$ & \text{\cdots \;}; 19$ \vspace{1mm}\\ 1.11. Center z(G_2) of G_2$ \quad & \text{\cdots \;}; 21$ \vspace{1mm}\\ 1.12. Complex exceptional Lie group G_2^C$ \quad & \text{\cdots \;}; 21$ \vspace{1mm}\\ 1.13. Non-compact exceptional Lie group G_{\{2(2)\}} of type G_2$ \quad & \text{\cdots \;}; 22$ \vspace{1mm}\\ \quad \quad \quad & \\ 1.14. Principle of triality in S(8) \quad & \text{\cdots \;}; 23$ \vspace{1mm}\\ 1.15. Spinor group Spin(7) \quad & \text{\cdots \;}; 26$ \vspace{1mm}\\ 1.16. Spinor group Spin(8) \quad & \text{\cdots \;}; 28$ \vspace{3mm} \end{tabular} $	(not rendered)	—
0902.0431_TAB_006	—	—	\begin{tabular}{lr} 2.1. Exceptional Jordan algebra $\text{\es {J}}$ \quad & \text{\cdots \;}; 31$ \vspace{1mm}\\ 2.2. Compact exceptional Lie group F_4$ \quad & \text{\cdots \;}; 33$ \vspace{1mm}\\ 2.3. Lie algebra \text{\es {f}}_4 of F_4$ \quad & \text{\cdots \;}; 36$ \vspace{1mm}\\ 2.4. Simplicity of \text{\es {f}}_4^C$ \quad & \text{\cdots \;}; 41$ \vspace{1mm}\\ 2.5. Killing form of \text{\es {f}}_4^C$ \quad & \text{\cdots \;}; 45$ \vspace{1mm}\\ 2.6. Roots of \text{\es {f}}_4^C$ \quad & \text{\cdots \;}; 47$ \vspace{1mm}\\ 2.7. Subgroup Spin(9) of F_4$ \quad & \text{\cdots \;}; 51$ \vspace{1mm}\\ 2.8. Connectedness of F_4$ \quad & \text{\cdots \;}; 55$ \vspace{1mm}\\ 2.9. Involution \sigma and subgroup Spin(9) of F_4$ \quad & \text{\cdots \;}; 57$ \vspace{1mm}\\ 2.10. Center z(F_4) of F_4$ \quad & \text{\cdots \;}; 58$ \vspace{1mm}\\ 2.11. Involution \gamma and subgroup (Sp(1) \times Sp(3))/\text{\es {Z}}_2 of F_4$ & \text{\cdots \;}; 59$ \vspace{1mm} \\ 2.12. Automorphism w of order 3 and subgroup (SU(3) \times SU(3))/\text{\es {Z}}_3 of F_4$ & \text{\cdots \;}; 63$ \vspace{1mm}\\ 2.13. Complex exceptional Lie group F_4^C$ \quad & \text{\cdots \;}; 65$ \vspace{1mm}\\ 2.14. Non-compact exceptional Lie groups F_{\{4(4)\}} and F_{\{4(-20)\}} of type F_4$ & \text{\cdots \;}; 66$; \\ \end{tabular} $	(not rendered)	—

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_TAB_007	—	—	\begin{tabular}{lr} 3.1. Compact exceptional Lie group $E_6 \sqquad \& \cdots \;$; 68\$ \vspace{1mm}\\ 3.2. Lie algebra $\text{\es {e}}_6$ of $E_6 \sqquad \& \cdots \;$; 68\$ \vspace{1mm}\\ 3.3. Simplicity of $\text{\es {e}}_6^C \sqquad \& \cdots \;$; 70\$ \vspace{1mm}\\ 3.4. Element $A \vee B$ of $\text{\es {e}}_6^C \sqquad \& \cdots \;$; 71\$ \vspace{1mm}\\ 3.5. Killing form of $\text{\es {e}}_6^C \sqquad \& \cdots \;$; 73\$ \vspace{1mm}\\ 3.6. Roots of $\text{\es {e}}_6^C \sqquad \& \cdots \;$; 75\$ \vspace{1mm}\\ 3.7. Involution τ and subgroup F_4 of $E_6 \sqquad \& \cdots \;$; 82\$ \vspace{1mm}\\ 3.8. Connectedness of $E_6 \sqquad \& \cdots \;$; 82\$ \vspace{1mm}\\ 3.9. Center $z(E_6)$ of $E_6 \sqquad \& \cdots \;$; 85\$ \vspace{1mm}\\ 3.10. Involution σ and subgroup $(U(1) \times \text{Spin}(10))/\text{\textit{Z}}_4$ of $E_6 \& \cdots \;$; 86\$ \vspace{1mm}\\ 3.11. Involution γ and subgroup $(\text{Sp}(1) \times \text{SU}(6))/\text{\textit{Z}}_2$ of $E_6 \& \cdots \;$; 90\$ \vspace{1mm}\\ 3.12. Involution $\tau\gamma$ and subgroup $\text{Sp}(4)/\text{\textit{Z}}_2$ of $E_6 \sqquad \& \cdots \;$; 95\$ \vspace{1mm}\\ 3.13. Automorphism w of order 3 and subgroup $(\text{SU}(3) \times \text{SU}(3) \times \sqquad \text{\textit{Z}}_3)$ of $E_6 \sqquad \& \cdots \;$; 100\$ \vspace{1mm}\\ 3.14. Complex exceptional Lie group $E_6^C \sqquad \& \cdots \;$; 105\$ \vspace{1mm}\\ 3.15. Non-compact exceptional Lie groups $E_{\{6(6)\}}, E_{\{6(2)\}}, E_{\{6(-14)\}}$ and $\sqquad E_{\{6(-26)\}}$ of type $E_6 \quad \& \cdots \;$; 105 \; \end{tabular}	(not rendered)	—
0902.0431_TAB_008	—	—	\begin{tabular}{lr} 4.1. Freudenthal vector space $\text{\es {P}}^C \sqquad \& \cdots \;$; 106\$ \vspace{1mm}\\ 4.2. Compact exceptional Lie group $E_7 \sqquad \& \cdots \;$; 108\$ \vspace{1mm}\\ 4.3. Lie algebra $\text{\es {e}}_7$ of $E_7 \sqquad \& \cdots \;$; 108\$ \vspace{1mm}\\ 4.4. Simplicity of $\text{\es {e}}_7^C \sqquad \& \cdots \;$; 114\$ \vspace{1mm}\\ 4.5. Killing form of $\text{\es {e}}_7^C \sqquad \& \cdots \;$; 116\$ \vspace{1mm}\\ 4.6. Roots of $\text{\es {e}}_7^C \sqquad \& \cdots \;$; 118\$ \vspace{1mm}\\ 4.7. Subgroups E_6 and $U(1)$ of $E_7 \sqquad \& \cdots \;$; 124\$ \vspace{1mm}\\ 4.8. Connectedness of $E_7 \sqquad \& \cdots \;$; 126\$ \vspace{1mm}\\ 4.9. Center $z(E_7)$ of $E_7 \sqquad \& \cdots \;$; 129\$ \vspace{1mm}\\ 4.10. Involution ι and subgroup $(U(1) \times E_6)/\text{\textit{Z}}_3$ of $E_7 \sqquad \& \cdots \;$; 130\$ \vspace{1mm}\\ 4.11. Involution σ and subgroup $(\text{SU}(2) \times \text{Spin}(12))/\text{\textit{Z}}_2$ of $E_7 \sqquad \& \cdots \;$; 133\$ \vspace{1mm}\\ 4.12. Involution $\tau\gamma$ and subgroup $\text{SU}(8)/\text{\textit{Z}}_2$ of $E_7 \& \cdots \;$; 142\$ \vspace{1mm}\\ 4.13. Automorphism w of order 3 and subgroup $(\text{SU}(3) \times \text{SU}(6))/\text{\textit{Z}}_3 \sqquad \quad \sqquad$ of $E_7 \quad \& \cdots \;$; 146\$ \vspace{1mm}\\ 4.14. Complex exceptional Lie group $E_7^C \sqquad \& \cdots \;$; 155\$ \vspace{1mm}\\ 4.15. Non-compact exceptional Lie groups $E_{\{7(7)\}}, E_{\{7(-5)\}}$ and $E_{\{7(-25)\}}$ $\sqquad \quad \sqquad$ of type $E_7 \sqquad \& \cdots \;$; 155 \; \end{tabular}	(not rendered)	—

Identifier	Page	Conf.	LaTeX source	Rendered	Image
0902.0431_TAB_014	—	—	\begin{tabular}{cccccccc} 1\$ & \$\longrightarrow\$ & & & & & & \\ \$Spin(11)\$ & \$\longrightarrow\$ & \$(E_7)^{\{\kappa,\mu\}}\$ & & & & & \\ & \$\longrightarrow\$ & \$S^{11}\$ & \$\longrightarrow\$ & & & & \\ & & \$\vspace{0.7mm}\$ & \$\downarrow\$ & \$p'\$ & & & \\ & & \$\downarrow\$ & \$p\$ & \$\downarrow\$ & \$=\$ & & \\ & & \$\vspace{0.7mm}\$ & \$1\$ & \$\longrightarrow\$ & \$S_0(11)\$ & & \\ & \$\longrightarrow\$ & \$S_0(12)\$ & \$\longrightarrow\$ & & & & \\ & \$S^{11}\$ & \$\longrightarrow\$ & & & & & \\ \end{tabular}	(not rendered)	—